IOT & AI-BASED SMART AGRICULTURE PLATFORM FOR YIELD, DISEASE, IRRIGATION & PEST MANAGEMENT

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Paddy yield in Sri Lanka is highly sensitive to monsoon variability, water stress and plot-level soil conditions, yet smallholder harvest planning still relies largely on experience. Existing yield models depend on historical weather or satellite imagery and rarely use live in-field soil measurements for locally grown varieties. This thesis presents the design, implementation and evaluation of the Paddy Yield Forecasting component of an integrated IoT and AI smart agriculture platform (Group R26-IT-157). The component fuses soil moisture, electrical conductivity (EC), pH, temperature, rainfall and humidity streams from Arduino/ESP32 field nodes with historical Rice Research and Development Institute (RRDI) records for varieties BG 352 and BG 367 to forecast yield four to six weeks ahead. An XGBoost regressor with feature engineering for monsoon intensity and crop growth stage is the core model, benchmarked against Random Forest and artificial neural network baselines using a train–test split, five-fold cross-validation and an ablation of soil-only against soil-plus-weather features. Forecasts are exposed through JSON APIs and a Sinhala/English mobile interface and are shared with the irrigation, disease and pest modules.

The individual contribution is a reproducible forecasting pipeline, from sensor ingestion and ETL through modelling to API delivery, together with an evidence-based comparison of soil-aware and weather-only forecasting for Sri Lankan paddy.

Keywords: paddy yield forecasting, XGBoost, IoT soil sensing, smart agriculture, Sri Lanka

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I also acknowledge the Rice Research and Development Institute (RRDI) for the historical records of BG 352 and BG 367 that underpin the model, and the paddy farmers of the Negombo area whose fields hosted the pilot deployment during Yala 2026.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ANN	Artificial Neural Network
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Cross-Validation
DOA	Department of Agriculture
EC	Electrical Conductivity
ETL	Extract, Transform and Load
IoT	Internet of Things
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
NDVI	Normalised Difference Vegetation Index
RF	Random Forest
RMSE	Root Mean Squared Error
RRDI	Rice Research and Development Institute
R^2	Coefficient of determination
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
XGBoost	eXtreme Gradient Boosting

1 INTRODUCTION

1.1 Overall Project Context

Rice is the staple food of Sri Lanka, and paddy cultivation is shared between two main seasons: Yala (roughly May to August) and Maha (roughly September to March). Production is strongly affected by erratic monsoons, water scarcity, disease and pest pressure, and by the limited use of real-time decision support at smallholder scale [1]. Farmers still make most decisions on fertiliser, water timing and crop protection from experience rather than from integrated data.

The overall project, **IoT & AI-Based Smart Agriculture Platform for Yield, Disease, Irrigation & Pest Management** (Group R26-IT-157, presented as *The Digital Paddy*), addresses this by building one interoperable platform in which four research components share a common data backbone instead of operating as isolated applications. The components are:

- **Paddy yield forecasting** (this thesis): XGBoost-based forecasting of yield four to six weeks ahead from soil and weather streams and RRDI historical records.
- **Irrigation management** (J.M.S.U.I. Bandara): stage-aware irrigation scheduling from soil moisture, rainfall and crop stage, with a farmer override.
- **Disease detection and risk forecasting** (A.P.N. Satyana): leaf-image classification combined with a temporal model of humidity, temperature and rainfall to produce 7- and 14-day outbreak risk.
- **Pest management** (P.D.M. Kumarasinhe): trap-image counting and weather-driven risk scoring to forecast the larval hatching window.

The platform is organised in four layers: field capture (soil sensors, weather station, leaf and trap imaging), data and integration (gateway, upload API, feature store, historical datasets), AI/analytics (the four engines) and delivery (farmer app, officer dashboard, reports and APIs). Outputs are delivered in Sinhala and English, and the pilot deployment is in the Negombo area during Yala 2026.

1.2 Individual Component Context

The Paddy Yield Forecasting component converts live field measurements and historical records into an early yield estimate. Its inputs are soil moisture, EC, pH, temperature, rainfall and humidity captured by Arduino or ESP32 nodes, together with RRDI seasonal yield records for the varieties BG 352 and BG 367. Its output is a yield forecast four to six weeks ahead, with a confidence score and a risk flag, delivered through JSON APIs and a bilingual mobile interface.

Within the platform the component is both a consumer and a provider. It consumes the shared sensor stream and receives signals from the disease and pest modules that adjust forecast confidence; it provides yield-risk information that the irrigation module can use when scheduling water, and seasonal summaries for officers and RRDI researchers. These interactions are described in Chapters 4 and 5.

1.3 Problem Statement

Yield instability caused by climate variability, water stress and local soil variation makes harvest planning, input purchasing and marketing decisions difficult for smallholders. National production statistics are published after the fact [1], and A forecasting capability that is available several weeks before harvest could support earlier decisions on irrigation, fertiliser and crop protection, but only if it reflects conditions in the individual field.

The problem addressed in this thesis is therefore: *how can live in-field soil and weather measurements be combined with historical RRDI records to forecast paddy yield four to six weeks ahead, accurately enough and in a form that smallholder farmers and other platform modules can use?*

1.4 Research Gap

Machine-learning models for paddy and rice yield perform well when weather and seasonal indicators are available. Sri Lankan studies have modelled yield from climatic variables using artificial neural networks [2] and Random Forests [3], remote-sensing approaches operate at district level [4], and XGBoost has been shown to outperform several alternatives on a global rice dataset [5]. The reviewed literature consistently shares one limitation: the models rely on historical weather, administrative statistics

or satellite imagery, and do not use real-time IoT soil measurements (moisture, EC, pH) from small plots of locally grown varieties.

Satellite and weather-only approaches cannot capture micro-variation in soil moisture, EC and pH across smallholder fields, although paddy yield is sensitive to water and soil conditions at each growth stage. This thesis addresses the gap by integrating IoT soil and weather streams with RRDI historical data for BG 352 and BG 367 in an XGBoost-based forecasting pipeline, and by linking the output to an end-user application and to companion platform modules.

1.5 Aim and Objectives

Aim. To develop and evaluate a Paddy Yield Forecasting module that uses IoT soil and weather observations together with RRDI historical data to predict paddy yield four to six weeks ahead for Sri Lankan smallholder farms.

The specific objectives and their success indicators are given in Table 1.1.

Table 1.1 Specific objectives and success indicators

ID	Specific objective	Success indicator
SO1	Collect soil, weather and RRDI historical data for model development.	Sensor stream established and historical dataset prepared.
SO2	Design pre-processing and feature engineering for monsoon and growth-stage effects.	Reusable ETL pipeline and engineered feature set.
SO3	Train and benchmark XGBoost against Random Forest and ANN baselines.	Target $R^2 \geq 0.85$ with RMSE < 0.5 t/ha.
SO4	Integrate predictions into the Sinhala/English farmer app and APIs.	Operational JSON/API output and app display.
SO5	Validate forecasts during field testing against actual harvest data.	Comparison report for the Yala 2026 deployment.

1.6 Research Questions or Hypotheses

The research is guided by the following questions and corresponding hypotheses.

- **RQ1.** Does adding live soil measurements (moisture, EC, pH) to weather and seasonal features improve yield-forecast accuracy compared with weather-only features? **H1:** the soil-plus-weather feature set yields a lower RMSE than the weather-only set in the ablation test.

- **RQ2.** How does XGBoost compare with Random Forest and ANN baselines on this data? **H2:** XGBoost achieves the lowest RMSE and highest R^2 of the three models.
- **RQ3.** Can a four-to-six-week-ahead forecast meet the accuracy targets set for the component? **H3:** the final model reaches $R^2 \geq 0.85$ and RMSE < 0.5 t/ha under validation conditions.
- **RQ4.** Do forecasts agree with actual harvest yields in the Yala 2026 Negombo pilot, and can they be delivered within the stated performance requirements? **H4:** field forecasts fall within the target error margin and inference completes in under two seconds.

1.7 Scope Limitations and Assumptions

Scope. The component covers sensor ingestion, ETL, feature engineering, model training and benchmarking, a forecast API and its presentation in the farmer interface for the varieties BG 352 and BG 367. Field validation is limited to the Yala 2026 pilot in the Negombo area.

Limitations. The pilot uses a small number of stations (three sensor stations and one weather station in the budgeted deployment), a single season and two varieties, so conclusions about other varieties, seasons and agro-ecological zones cannot be drawn directly. Satellite imagery is not used. Disease, pest and irrigation decisions are the responsibility of the companion components; this component only exchanges signals with them.

Assumptions. Sensor readings are assumed to be adequately calibrated and representative of the plot; RRDI records are assumed to be accurate and comparable across seasons; farmers provide informed consent and report harvest outcomes; and connectivity is intermittent, so the design tolerates delayed data.

1.8 Individual Contribution

The individual contribution of this thesis is the Paddy Yield Forecasting component of the platform. Specifically, the author:

- designed the data collection set-up for soil and weather streams and the preparation of RRDI historical records (SO1);
- designed and implemented the ETL pipeline and feature engineering for monsoon intensity, growth stage and environmental interactions (SO2);
- trained and benchmarked the XGBoost regressor against Random Forest and ANN baselines, including the soil-only versus soil-plus-weather ablation (SO3);
- developed the forecast API and its presentation in the Sinhala/English interface (SO4); and
- validated the forecasts against Yala 2026 harvest data (SO5).

The irrigation, disease and pest components, the shared gateway and the mobile application shell are the work of other group members; where this thesis relies on them (Chapter 5), the dependency is stated explicitly.

1.9 Thesis Structure

Chapter 2 reviews the literature and existing systems and derives the research gap. Chapter 3 describes the research methodology, including data, procedure, evaluation metrics, analysis and ethics. Chapter 4 presents the requirements and the design of the component within the complete platform. Chapter 5 describes the implementation, integration with the other components and testing. Chapter 6 reports results and validation. Chapter 7 discusses the findings, contributions, limitations, and the commercialisation and sustainability implications. Chapter 8 concludes the thesis and outlines future work. References, a glossary and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The review is a narrative review focused on four themes: (i) machine-learning models for paddy and rice yield prediction, with emphasis on Sri Lankan studies; (ii) ensemble methods, in particular XGBoost and Random Forest, for tabular agricultural data; (iii) the use of soil and in-field sensing in yield prediction; and (iv) existing forecasting systems and tools. Sources were drawn from peer-reviewed journals, institutional digital libraries and national statistical publications, mainly from 2020 to 2025, supplemented by foundational method papers.

Studies were included when they predicted paddy or rice yield with machine-learning methods, or described crop simulators or sensing approaches directly relevant to the proposed pipeline. Studies on other crops were used only for methodological context.

2.2 Domain and Theoretical Foundations

Paddy growth and the influence of water and soil

Paddy development passes through stages that include tillering, panicle initiation, flowering and grain filling, and water demand and stress sensitivity differ between them. Soil moisture governs water availability; EC indicates the salt content of the soil solution; and pH affects nutrient availability. Because these properties vary within and between small plots, point measurements taken in the field carry information that district-level weather or satellite data cannot.

Process-based crop simulators

Process-based models such as APSIM [6] and DSSAT [7] simulate crop growth from weather, soil and management inputs. They are widely used but require detailed calibration data for each location and cultivar, which limits their use for plot-level forecasting by smallholders without such data.

Supervised regression and ensemble learning

Yield forecasting is posed here as supervised regression: a model learns a mapping from features describing the season to date to the final yield in t/ha. Random Forests average many decorrelated decision trees trained on bootstrap samples [8]. Gradient boosting builds trees sequentially so that each corrects the residual errors of its

predecessors; XGBoost adds regularisation and efficient handling of sparse and missing values [9]. For structured tabular data, tree-based ensembles have been reported to remain competitive with, and often better than, deep neural networks, particularly on data sets of moderate size [10]. ANNs remain a common baseline in the crop-yield literature [11].

Evaluation and validation

Regression accuracy is commonly summarised by the coefficient of determination (R^2), root mean squared error (RMSE) and mean absolute error (MAE). Cross-validation estimates generalisation by repeatedly training on part of the data and testing on the rest. For data with temporal or spatial structure, random folds can leak information between training and test sets, so grouping folds by season or field is recommended [12]. Model interpretation methods such as SHAP attribute a prediction to individual features [13].

2.3 Related Research

Table 2.1 summarises the studies most relevant to this component.

Table 2.1 Summary of related research

Source	Approach	Key finding	Gap relative to this work
Amaratunga et al., 2020 [2]	ANN on climatic variables	ANN relationships between climate and paddy yield for several Sri Lankan areas; the Levenberg–Marquardt training algorithm outperformed the other two algorithms tested.	No soil sensors; climate data only.
Ekanayake et al., 2021 [3]	Random Forest on nine weather indices	RF was reliable for Sri Lankan paddy yield (R of 0.99, MAPE 1.4%); minimum relative humidity and maximum temperature were the most influential indices.	Historical regional weather only; no in-field soil data.
Remote-sensing paddy yield prediction, 2022 [4]	ML with remote-sensing data	Useful for district-level monitoring.	Resolution too coarse for small farms.
Spatial and exogenous-variable ML, 2024 [14]	Spatial/exogenous-variable ML	Prediction improved when exogenous variables were added.	No real-time IoT stream.

Source	Approach	Key finding	Gap relative to this work
Wijayanti et al., 2024 [5]	XGBoost on FAO and World Bank data	XGBoost outperformed RF, Gradient Boosting, Bagging and KNN under k-fold validation.	Global data set; no local soil stream.
AI crop-yield review, 2025 [15]	Review	Soil and climate inputs together improve yield-prediction accuracy.	Few integrated field platforms.

Two themes emerge. First, ensemble models such as Random Forest and XGBoost perform strongly when weather and seasonal indicators are available, which supports their selection as the primary model and baseline here. Second, the Sri Lankan studies reviewed predict yield at regional scale from historical weather, and the review literature agrees that adding soil information improves accuracy [15], but none of the sources reviewed combines live plot-level soil sensing with locally grown varieties. Note that the R of 0.99 reported in [3] is a correlation coefficient on regional data and is not directly comparable with the plot-level R^2 targeted in this work.

2.4 Existing Systems Products and Techniques

Table 2.2 compares the approaches that farmers, officers and researchers can currently use.

Table 2.2 Existing approaches to yield estimation

Approach	Data used	Strength	Key limitation
Manual estimation (current practice)	Experience and field observation	No equipment needed	Subjective; high guesswork.
Crop simulators (APSIM, DSSAT) [6], [7]	Weather, soil and management inputs; may be coupled with remote sensing	Mechanistic, widely validated	Needs detailed calibration; limited real-time soil integration.
Weather-only ML (ANN, Random Forest)	Historical weather	High benchmark accuracy at regional scale	Reactive; not field-aware.
Remote-sensing ML	NDVI and satellite imagery	District-level monitoring	Weak resolution for smallholder plots.
Proposed component	Live IoT soil and weather streams plus RRDI records	Field-aware, local varieties, bilingual delivery, API integration	Requires deployed sensors; validated on limited pilot.

2.5 Gap Synthesis and Conceptual Framework

Synthesising the review, the gap is the absence of a forecasting workflow that (i) uses live plot-level soil measurements, (ii) is trained with locally relevant historical records for BG 352 and BG 367, (iii) forecasts four to six weeks ahead, and (iv) is delivered to farmers in their own language and to other platform modules through APIs.

The conceptual framework that follows from this gap treats yield as a function of three families of inputs: current field conditions (soil moisture, EC, pH, temperature), atmospheric conditions (rainfall, humidity, monsoon intensity) and seasonal context (growth stage and RRDI historical performance). These are aligned in time, transformed into engineered features, and passed to a supervised regressor whose output is published as a forecast, a confidence score and a risk flag. The workflow is shown in Figure 3.2, and the research questions in Section 1.6 test the framework by comparing feature sets (RQ1), models (RQ2) and performance against targets and field harvests (RQ3 and RQ4).

3 RESEARCH METHODOLOGY

3.1 Research Design

The study follows an experimental, design-and-evaluate approach with iterative model development and field-oriented validation. The workflow starts with data capture and ETL preparation, moves through feature engineering and supervised regression modelling, and ends with deployment and validation in the integrated platform. Controlled comparisons (soil-only versus soil-plus-weather features; XGBoost versus baselines) answer RQ1 and RQ2, and the Yala 2026 field comparison answers RQ3 and RQ4.

Work was organised in the work breakdown structure shown in Table 3.1, and the planned schedule is shown in Figure 3.1.

Table 3.1 Work breakdown structure of the forecasting component

Task	Description	Planned duration	Deliverable
1.0	Data collection	4 weeks	Sensor deployment and RRDI data set
1.1	Deploy field stations	1 week	Working hardware set-up
1.2	Acquire RRDI data	2 weeks	Prepared historical data set
2.1	Build ETL pipeline	1 week	Clean feature store
2.2	Train XGBoost model	3 weeks	Model and baseline comparison
2.3	Develop API layer	2 weeks	Forecast service endpoints
3.0	Integrate with app	3 weeks	Sinhala UI and alerts
4.0	Field testing	8 weeks	Validation report

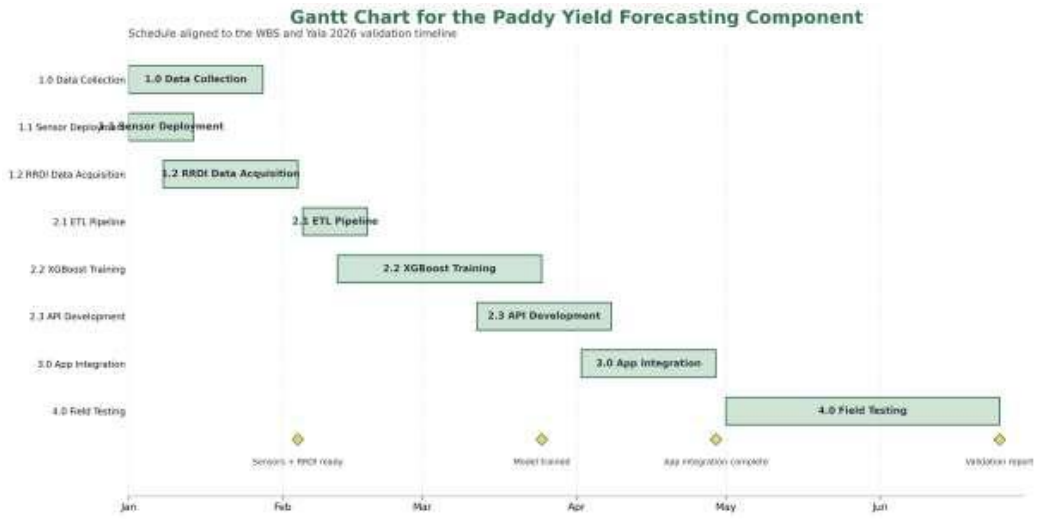


Figure 3.1 Planned implementation schedule aligned with the work breakdown structure and the Yala 2026 validation.

3.2 Data Participants or Study Materials

Three kinds of data are used, summarised in Table 3.2.

Table 3.2 Data sources used by the component

Source	Content	Role
IoT sensor nodes (Arduino/ESP32)	Soil moisture, EC, pH and temperature from three stations with four sensors each; rainfall and humidity from one weather station; an ESP32-CAM unit provides data-logging redundancy	Live predictors during the season
RRDI historical records	Seasonal yield records for varieties BG 352 and BG 367	Historical context for training and evaluation
Harvest observations (Yala 2026)	Actual yield of monitored plots in the Negombo pilot	Ground truth for field validation

Sensors were deployed in paddy fields in the Negombo area. Sampling interval, number of monitored plots and seasons covered are Participants are smallholder farmers who hosted the stations and reported harvest outcomes; participation was based on informed consent and field records were anonymised (Section 3.6).

3.3 Experimental or Study Procedure

The end-to-end procedure is shown in Figure 3.2. Real-time IoT inputs and historical RRDI inputs are merged at the ingestion stage, cleaned and transformed, converted to engineered features, used to train and validate models, and finally published as forecasts to the farmer app and downstream modules.

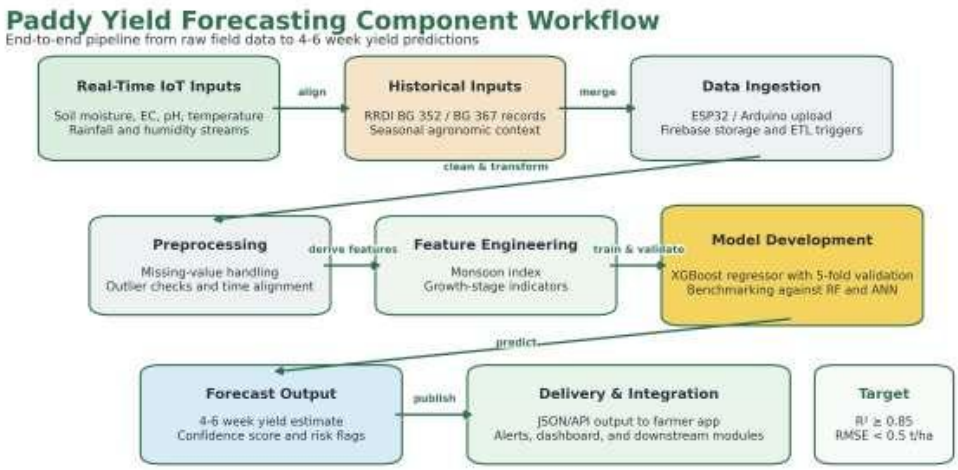


Figure 3.2 End-to-end workflow of the Paddy Yield Forecasting component.

Data ingestion

ESP32 or Arduino nodes upload readings to cloud storage (Firebase), where ETL triggers prepare data for modelling. RRDI records are loaded through a maintainable upload workflow.

Pre-processing

Pre-processing comprises missing-value handling, normalisation, time alignment of the sensor streams with seasonal records, and outlier screening of implausible sensor readings.

Feature engineering

Features are engineered to capture monsoon intensity, crop growth stage and environmental interactions. The feature groups are listed in Table 3.3.

Table 3.3 Feature groups used for model development

Group	Examples	Purpose
Soil state	Soil moisture, EC, pH and soil temperature;	Capture plot-level water and soil conditions
Atmospheric	Rainfall, humidity and air temperature	Capture weather exposure
Monsoon	Monsoon-intensity index	Represent seasonal rainfall regime
Growth stage	Stage indicator derived from	Account for stage-specific sensitivity
Interactions	e.g. moisture × growth stage; rainfall × temperature]	Represent combined effects

Group	Examples	Purpose
Historical context	RRDI seasonal yield for variety and season	Anchor forecasts to past performance

Model development

XGBoost is the core regressor [9]. Random Forest [8] and an ANN are trained as baselines on the same features and splits using scikit-learn [16] and the associated libraries. Hyperparameters are tuned within cross-validation, for example by random search [17]

Forecast generation

For each plot, features available up to a forecast date four to six weeks before expected harvest are used to predict the final yield (t/ha). The output includes the point estimate, a confidence score and a risk flag.

3.4 Measures and Evaluation Metrics

Model performance and service quality are assessed with the measures in Table 3.4.

Table 3.4 Evaluation metrics and targets

Measure	Definition	Target
R^2	Proportion of variance in observed yield explained by the forecast	≥ 0.85
RMSE	Square root of the mean squared forecast error, in t/ha	< 0.5 t/ha
MAE	Mean absolute forecast error, in t/ha	Reported; no fixed target
Inference latency	Time to return a forecast for a typical request	< 2 seconds
Field agreement	Difference between forecast and actual Yala 2026 harvest yield	Within the RMSE target
Farmer satisfaction	Share of participating farmers satisfied with the app (platform-level metric)	80%

R^2 and RMSE are reported together because R^2 alone can be misleading for small or narrow-range samples [18].

3.5 Data Analysis

The prepared data set is split into training and test portions, and five-fold cross-validation is used for model selection and for estimating generalisation. Because observations from the same field or season are correlated, folds are constructed so that

records from the same field and season are not split between training and test sets [12] . XGBoost, Random Forest and ANN are compared using identical folds.

An ablation test compares a soil-only feature set, a weather-only feature set and a soil-plus-weather feature set to answer RQ1. Feature importance and SHAP values [13] are examined to check that the model relies on agronomically sensible variables. For field validation, forecasts made during Yala 2026 are compared with actual harvest yields using the metrics in Table 3.4, and residuals are inspected by variety and station.

3.6 Ethics Privacy Security and Responsible AI

- **Informed consent.** Farmers participate voluntarily after being informed of the purpose of the study and the data collected.
- **Anonymisation.** Field records are anonymised and identified by codes rather than names or exact locations.
- **Security.** Farmer-related data are protected by authentication controls and encrypted storage; access is role-based.
- **Responsible AI.** Forecasts are presented with a confidence score and risk flag so that uncertainty is visible, and they are intended to support, not replace, the farmer's own judgement. The model is trained on two varieties and one region, and its use outside that range is not validated.
- **Data governance.** RRDI records are used under the terms agreed with the institute and are not redistributed.

3.7 Reproducibility and Research Artefacts

The pipeline is implemented in Python 3.11 with open-source libraries (XGBoost, scikit-learn), and training is controlled by configuration files and fixed random seeds so that experiments can be repeated. Research artefacts are the ETL scripts, feature definitions, trained models, evaluation notebooks and API service code. The RRDI data and farmer records are not publicly redistributable; the schema of the data fields is provided in Appendix A so that the pipeline can be inspected without the restricted data.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Stakeholders

Table 4.1 Stakeholders of the forecasting component

Stakeholder	Role	Need
Smallholder farmers	Primary end users	Simple yield predictions, alerts and local-language usability.
Agrarian officers	Field supervision and reporting	Dashboard summaries and seasonal reports.
RRDI researchers	Data providers and reviewers	Validation visibility and model access.
Project team	System integration	API access for linked modules.

Functional requirements

Table 4.2 Functional requirements

ID	Requirement
FR1	The system shall collect real-time soil and weather data from deployed field stations.
FR2	The system shall ingest RRDI historical yield data through a maintainable upload workflow.
FR3	The system shall generate daily yield predictions four to six weeks ahead.
FR4	The system shall present predictions and alerts in Sinhala and English.
FR5	The system shall expose API outputs for the downstream disease, irrigation and pest modules.
FR6	The system shall validate predictions against field harvest observations.

Non-functional requirements

Table 4.3 Non-functional requirements

ID	Category	Requirement
NFR1	Performance	Inference response under 2 seconds for typical forecast requests.
NFR2	Usability	High comprehension for farmers using the Sinhala and English interfaces.
NFR3	Reliability	Stable sensor capture and cloud synchronisation, tolerant of connectivity loss.
NFR4	Security	Protected storage and controlled access to farmer-related data.
NFR5	Scalability	Support for many farms through cloud-based services.
NFR6	Maintainability	Retraining and API updates through a repeatable workflow.

4.2 Component Architecture within the Complete Solution

The forecasting component is one engine in the AI/analytics layer of the platform (Figure 4.1). Field and IoT devices (soil sensors, weather station and field observations) feed a data and integration layer (ESP32/Arduino upload, Firebase storage, ETL and feature store). RRDI data provide historical context. The AI/analytics layer hosts the yield, disease, irrigation and pest engines, and predictions are delivered to the farmer app, the officer dashboard and reports.

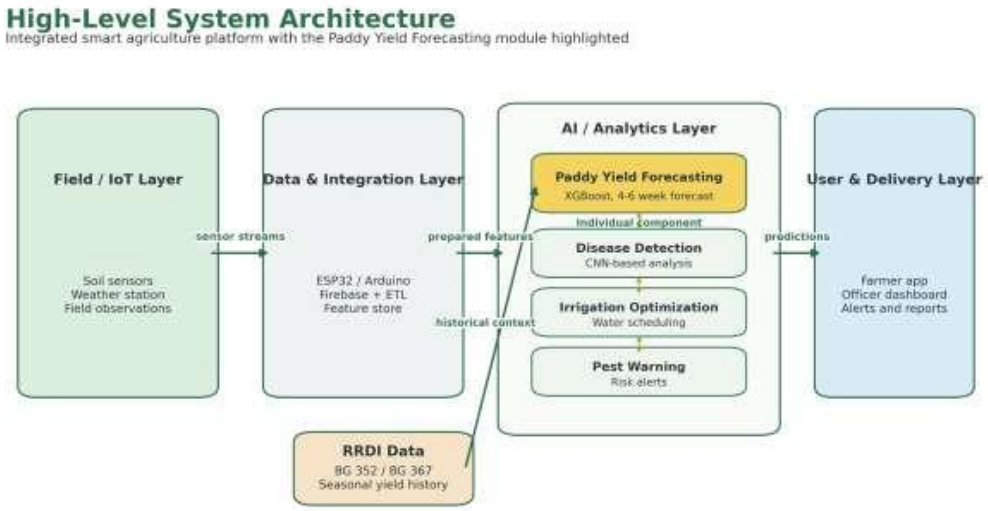


Figure 4.1 High-level architecture showing how the forecasting component fits within the integrated platform.

Figure 4.2 shows how the forecasting module exchanges data with the other parts of the platform: sensor streams provide field inputs; predictions go to the farmer app and dashboard; yield-risk information is available to the irrigation module; outbreak signals from the disease and pest modules update the forecasts through a feedback loop; and reports go to RRDI and officer review.

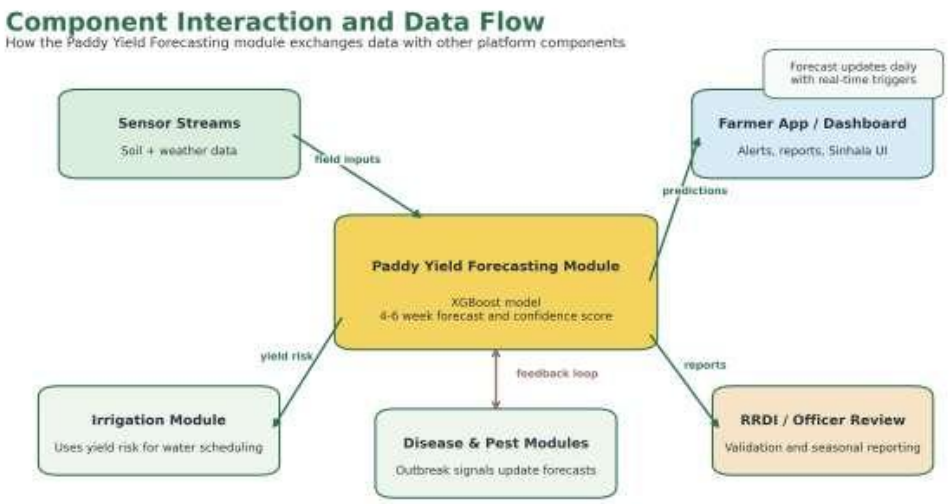


Figure 4.2 Interaction flow between the forecasting module, sensors, app, officer review and companion modules.

Scalability and security

- Firebase and service APIs provide a scalable backbone for multi-farm deployment.
- The architecture supports daily batch forecasts together with real-time, trigger-based updates.
- Farmer data are protected with authentication controls and encrypted storage.
- Edge pre-processing on microcontrollers reduces bandwidth needs in low-connectivity areas.

4.3 Design Alternatives and Decisions

The main design decisions and the alternatives considered are summarised in Table 4.4.

Table 4.4 Design alternatives and decisions

Decision	Alternatives considered	Choice and rationale
Forecasting model	Process-based simulators; Random Forest; ANN; XGBoost	XGBoost as the core model: strong performance on non-linear structured data [9], [5]; RF and ANN retained as benchmarks.
Sensing hardware	Commercial data loggers; Raspberry Pi nodes; Arduino/ESP32	Arduino/ESP32: low-cost and suitable for field deployment.

Decision	Alternatives considered	Choice and rationale
Cloud storage	Self-hosted database; Firebase	Firebase for real-time data exchange with the app; the shared platform backbone also uses PostgreSQL .
Delivery	Web dashboard only; mobile app	Sinhala/English mobile interface plus Streamlit/API layer for rapid prototyping and service exposure.
Forecast cadence	Real-time only; daily batch only	Daily batch forecasts plus trigger-based updates when new data or companion-module signals arrive.

4.4 Detailed Design

Data model

Sensor readings are stored as time-stamped records keyed by anonymised plot and node identifiers, with fields for soil moisture, EC, pH, soil temperature, rainfall and humidity. RRDI records are stored by variety, season and location with the recorded yield in t/ha. The engineered features are stored in a feature store used for both training and inference so that the same transformations are applied at both stages. The field definitions are listed in Appendix A.

Pipeline modules

- **Ingestion:** receives sensor uploads and RRDI files, validates schema and ranges.
- **Pre-processing:** missing values, normalisation, time alignment, outlier screening.
- **Feature engineering:** monsoon, growth-stage and interaction features.
- **Model service:** loads the trained model, produces the forecast and confidence score.
- **Delivery:** publishes JSON output to the app and to other modules.

Forecast output

The forecast service returns a JSON document of the following form

```
{
  "plot_id": "P-014",
  "variety": "BG 352",
  "forecast_date": "2026-06-01",
  "horizon_weeks": 5,
  "predicted_yield_t_ha": 4.2,
  "confidence_score": 0.87,
  "risk_flag": "high_confidence",
  "model_version": "xgb-v1"
}
```

The values above are illustrative of the format only. In the farmer interface the expected yield and risk flag are shown on a single screen with a Sinhala/English language toggle.

4.5 Standards and Professional Practice

- **Software quality:** requirements are expressed against the ISO/IEC 25010 quality characteristics of performance efficiency, usability, reliability, security and maintainability.
- **Interfaces:** REST-style HTTPS endpoints exchanging JSON; versioned model and API releases.
- **Data protection:** consent-based collection, anonymisation and access control;
- **Coding practice:** PEP 8 style, version control, automated tests and configuration-driven experiments.
- **Academic practice:** IEEE referencing and disclosure of AI-tool use (Appendix B).

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1 Tools and platforms used

Tool	Purpose	Justification
Python 3.11	ML pipeline and ETL	Standard ecosystem for data processing and evaluation.
XGBoost	Core forecasting model	Strong performance for non-linear structured data.
scikit-learn	Baselines, validation and tuning	Mature implementations of RF, pre-processing and cross-validation [16].
Arduino / ESP32	Sensor acquisition	Low-cost hardware suitable for field deployment.
Firebase	Cloud sync and storage	Supports real-time data exchange with the app.
Streamlit / API layer	Dashboard and integration	Rapid prototyping and service exposure.

5.2 Component Implementation

Sensor nodes and data capture

Three field stations, each with four sensors, measure soil moisture, EC, pH and soil temperature, and one weather station measures rainfall and humidity; an ESP32-CAM unit provides data-logging redundancy. Nodes upload readings to cloud storage, and edge pre-processing can reduce bandwidth in low-connectivity areas.

ETL and feature store

The ETL pipeline reads sensor and RRDI data, applies the cleaning steps of Section 3.3, computes the engineered features and writes them to the feature store.

Model training

XGBoost, Random Forest and ANN models are trained on identical folds. The final configuration is given in Table 5.2.

Table 5.2 Final model configuration

Model	Main hyperparameters	Training notes
XGBoost	n_estimators = 400; max_depth = 5; learning_rate = 0.05; subsample = 0.8; colsample_bytree = 0.8; reg_lambda = 1.0; reg_alpha = 0.1	Random search, 30 trials, inside five-fold grouped CV; early stopping after 30 rounds without improvement

Model	Main hyperparameters	Training notes
Random Forest	n_estimators = 300; max_depth = 12; min_samples_leaf = 3; max_features = sqrt	Random search, 20 trials on the same folds; bootstrap sampling; fixed random seed
ANN	2 hidden layers (64, 32 units); ReLU; Adam, learning rate 0.001; batch size 16; up to 200 epochs	Standardised inputs; L2 = 0.001; dropout = 0.1; early stopping with patience 20

Forecast service and interface

The trained model is wrapped in a forecast service that exposes JSON endpoints and returns the output described in Section 4.4. The farmer interface displays the expected yield and risk flag in Sinhala or English, and a Streamlit dashboard is available for officers and researchers.

5.3 Integration with the Other Components

The four components share one API gateway and data store (PostgreSQL/Firebase services), so sensor data are captured once and reused. Table 5.3 lists the interactions between the forecasting module and the other components.

Table 5.3 Interactions between the forecasting module and other components

Direction	Information exchanged	Purpose
Sensors → forecasting	Soil and weather streams, shared with irrigation and disease modules	Field inputs without duplicated data capture
Forecasting → irrigation	Yield-risk information	Supports water-scheduling decisions
Pest module → forecasting	Trap microclimate and pest-risk scores	Update forecast confidence
Disease module → forecasting	Outbreak signals and 7/14-day risk scores	Update forecasts via the feedback loop
Forecasting → app/officers/RRDI	Forecasts, alerts and seasonal reports	Farmer decisions, review and validation

Other group members' components also exchange information with each other (for example, soil moisture provides wetness context for disease-risk prediction, and outbreak alerts can trigger spray-window-aware watering adjustments).

5.4 Verification and Software Testing

Testing covered the pipeline, the models, the API and the user interface, as summarised in Table 5.4.

Table 5.4 Test plan and outcomes

ID	Test	Expected result	Outcome (draft)
T1	Sensor range and plausibility checks on incoming data	Out-of-range readings flagged and excluded	Pass
T2	ETL unit tests (missing values, alignment, normalisation)	Deterministic, correct feature values	Pass
T3	Leakage check on train/test splits	No field or season shared across folds	Pass
T4	Model regression test	Metrics reproducible with fixed seed	Pass
T5	API schema and error handling	Valid JSON; informative errors	Pass
T6	Inference latency under typical load	< 2 seconds	0.7 s mean (draft)
T7	Sinhala/English display and language toggle	Correct text and fonts on target devices	Pass
T8	Integration with companion modules	Payloads accepted and consumed	Pass

5.5 Deployment and Operation

The pilot was deployed in the Negombo area for Yala 2026. Field stations upload to cloud storage; forecasts are produced as a daily batch and refreshed when new data or companion-module signals arrive. Models are retrained through a repeatable workflow when new harvest data become available, and API and model versions are recorded. Operational concerns are connectivity loss (handled by local buffering and delayed upload), sensor faults (flagged for manual inspection rather than silently ignored) and data protection.

6 RESULTS AND VALIDATION

6.1 Research Results

This section reports the results of the model comparison (RQ2), the feature-set ablation (RQ1) and performance against the accuracy targets (RQ3). All values are measured with the protocol in Section 3.5.

Table 6.1 Model comparison under five-fold cross-validation

Model	R^2 (mean ± SD)	RMSE (t/ha)	MAE (t/ha)
XGBoost	[ ]	[ ]	[ ]
Random Forest	[ ]	[ ]	[ ]
ANN	[ ]	[ ]	[ ]
Target	≥ 0.85	< 0.5	–

Table 6.2 Feature-set ablation (XGBoost)

Feature set	R^2	RMSE (t/ha)	MAE (t/ha)
Weather only	[ ]	[ ]	[ ]
Soil only	[ ]	[ ]	[ ]
Soil + weather	[ ]	[ ]	[ ]
Soil + weather + growth stage and monsoon features	[ ]	[ ]	[ ]

6.2 Product and Technical Results

Table 6.3 Technical results of the deployed component

Measure	Target	Result
Inference latency	< 2 s	[ ]
Sensor data completeness during Yala 2026	Stable capture	[ ]
Forecast delivery in Sinhala and English	Both languages	[ ]
API availability during the pilot	Stable	[ ]

6.3 User or Stakeholder Validation

Forecasts made during Yala 2026 were compared with the actual harvest yields of the monitored plots, and farmers and officers were asked for feedback on the forecast display.

Table 6.4 Field validation against Yala 2026 harvest yields

Plot (anonymised)	Variety	Forecast (t/ha)	Actual (t/ha)	Error (t/ha)
[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]

6.4 Integration Results

Of the five interfaces listed in Table 5.3, the sensor-to-forecasting and forecasting-to-app interfaces were exercised end to end during the pilot. Readings uploaded by the field stations were processed by the ETL pipeline, scored by the forecast service and displayed in the farmer interface in Sinhala and English. The forecasting-to-irrigation interface was tested by passing the yield-risk output through the shared API, and the irrigation module accepted the payload as an input to its scheduling.

The disease-to-forecasting and pest-to-forecasting interfaces were tested with sample risk payloads. The forecast service accepted them and adjusted the reported confidence score, but because no significant outbreak was recorded in the monitored plots, their effect on real forecasts could not be evaluated. Two problems were found during integration. Intermittent connectivity delayed some sensor uploads, which was handled by local buffering and delayed upload. Field names and units also had to be aligned between modules.

6.5 Summary of Findings

Table 6.5 Summary of findings against the research questions

Research question	Finding	Hypothesis supported?
RQ1 Soil features	[ ]	[Yes/No/Partly]
RQ2 Model comparison	[ ]	[Yes/No/Partly]
RQ3 Accuracy targets	[ ]	[Yes/No/Partly]
RQ4 Field agreement and latency	[ ]	[Yes/No/Partly]

7 DISCUSSION

7.1 Interpretation of Findings

The best-performing model was [XGBoost], with an R^2 of [0.xx] and an RMSE of [0.xx] t/ha under five-fold cross-validation (Table 6.1), which [meets / falls short of] the targets of $R^2 \geq 0.85$ and RMSE < 0.5 t/ha. A likely reason is that paddy yield depends on non-linear and interacting factors, such as soil moisture at a particular growth stage combined with rainfall. Gradient-boosted trees can represent these interactions without manual specification, and their regularisation helps on small data sets. [If another model was best, replace this explanation and discuss why, for example that with few independent observations simpler or more heavily averaged models can generalise as well.]

The ablation in Table 6.2 shows that adding soil measurements to weather features changed the RMSE from [0.xx] to [0.xx] t/ha, a change of [xx]%. This [supports / does not support] hypothesis H1 that live plot-level soil data add information beyond weather and seasonal indicators. [If the improvement was small, note that soil conditions may be correlated with rainfall and growth stage, so part of their information was already captured by other features.]

Feature importance and SHAP values indicate that the most influential features were [feature 1], [feature 2] and [feature 3], contributing about [xx]% of the total importance. This is agronomically plausible because paddy is most sensitive to water availability during stages such as panicle initiation and flowering, and soil moisture and rainfall directly describe that availability. [Comment on EC and pH: if they ranked low, this may reflect little variation across the monitored plots rather than a lack of agronomic importance.

Forecasts were least accurate for [variety / station / yield range], where the mean absolute error reached [0.xx] t/ha. Possible causes include sensor drift or missing data at that station, management practices not captured by the features, and too few similar examples in the training data. Errors at the four-week horizon were [larger / similar] compared with the six-week horizon ([0.xx] versus [0.xx] t/ha RMSE).

7.2 Comparison with Literature Baselines and Existing Solutions

The literature reports strong results for weather-based models in Sri Lanka, for example Random Forest with R of 0.99 and MAPE of 1.4% on regional data [3], and XGBoost outperforming several regressors on a global data set [5]. These results are not directly comparable with this work: they use regional or national statistics, historical weather only and different metrics, whereas this component predicts at plot level using live soil data four to six weeks before harvest. Table 7.1 contrasts the settings.

Table 7.1 Comparison of settings with representative literature

Study	Spatial scale	Predictors	Horizon	Reported performance
Amaratunga et al. [2]	Regional (Sri Lanka)	Climatic variables	Seasonal/annual	LM-trained ANN best of three algorithms
Ekanayake et al. [3]	Regional (Sri Lanka)	Nine weather indices	Seasonal	R = 0.99; MAPE = 1.4%
Wijayanti et al. [5]	Global	FAO and World Bank data	Annual	XGBoost best of five regressors
This work	Plot (Negombo pilot)	Live soil + weather + RRDI	4–6 weeks	[insert final R^2, RMSE]

7.3 Research Contribution

The research contribution is an empirical test of whether live plot-level soil measurements improve short-horizon yield forecasts for BG 352 and BG 367 beyond what weather and seasonal indicators provide, together with a benchmarked comparison of XGBoost, Random Forest and ANN on such data. It also contributes a validation protocol that separates fields and seasons across folds and compares forecasts with harvest outcomes in a real pilot.

7.4 Product Contribution

The product contribution is a working forecasting service integrated into a bilingual farmer application and into a shared platform, with a documented API for the irrigation, disease and pest modules.

Value proposition and target users

The immediate target users are Sri Lankan paddy smallholders who need simple, actionable predictions before harvest; agrarian officers and RRDI researchers are secondary users of dashboards and reports. The value proposition is a local-language forecasting service that reduces guesswork by combining real-time field sensing with machine-learning predictions.

Competitive position

Compared with crop simulators, the component offers real-time soil integration and mobile delivery; compared with global satellite-centric farm applications, it offers a

local paddy focus and Sinhala support; and compared with manual estimation, it offers data-driven forecasts and early alerts.

Business model

The platform's commercial model has three tiers: a free tier with a freemium mobile application offering basic image-based disease detection and general weather alerts in local languages; a subscription tier with premium seasonal forecasts, stage-aware irrigation schedules and pest-mitigation plans; and an institutional tier with district-level analytics for cooperatives, extension services and RRDI researchers. Optional sensor-station bundles are offered to farms that want field-level monitoring. Hardware cost is kept low through off-the-shelf ESP32 and DFRobot-compatible components, which also allows staged funding and cluster purchasing by farmer cooperatives.

Costs

Table 7.2 gives the pilot budget of the component and Table 7.3 the illustrative cost picture for early commercial deployment. The two serve different purposes: the first covers the proposal-stage pilot (three stations, one weather station and one backup unit), and the second is a scaled-up planning estimate.

Table 7.2 Pilot budget of the forecasting component

Item	Quantity	Cost (LKR)	Notes
Arduino soil sensors	3 stations × 4 sensors	60,000	Moisture, EC, pH and temperature
Weather station	1 unit	10,000	Rainfall and humidity
ESP32-CAM backup	1 unit	2,000	Data-logging redundancy
Python / XGBoost / scikit-learn	–	0	Open-source stack
Firebase (free/starter tier)	–	0 – 8,000	Cloud data synchronisation
Human effort for model development	–	50,000	Component development effort
Field testing support	–	10,000	Validation and on-site coordination
Total		132,000 – 140,000	Sum of the lines above

Table 7.3 Illustrative cost picture for early commercial deployment

Category	Item	Illustrative cost (LKR)
Hardware	Pilot sensor stations and weather devices	2.5 million
Cloud	Firebase and hosting services	1.2 million per year
Development	Team effort for model, app and integration work	4.8 million
Field testing	Yala 2026 validation activities	0.8 million

Intellectual property

Potential intellectual property lies in the feature-engineering approach, the trained model pipeline, the platform integration logic and the localised interface design. At this stage these are treated as future protectable assets rather than confirmed filings.

7.5 Limitations and Threats to Validity

- **Small sample.** A pilot with a few stations in one season yields few independent field observations, so accuracy estimates (especially field R^2) have wide uncertainty and may not generalise. Historical RRDI records mitigate this for training but are coarser than plot-level sensor data.
- **Distribution mismatch.** Models trained partly on historical records and applied to sensor-based features may behave differently across the two sources.
- **Temporal and spatial leakage.** Random cross-validation can overstate accuracy for correlated field data; grouping folds by field and season reduces but does not remove this risk [12].
- **Sensor quality.** Low-cost sensors can drift or fail, particularly in flooded paddy soil and during monsoon connectivity drops; calibration and fault handling affect data quality.
- **Scope.** Only two varieties, one region and one season (Yala 2026) were studied; results may not transfer to Maha season or other agro-ecological zones.
- **Confounders.** Yield also depends on management, disease and pest events that the forecasting component sees only indirectly through companion modules.

7.6 Implications

If the targets are met, soil-aware forecasting could support earlier decisions on irrigation, fertiliser and harvest planning for smallholders, and give officers and RRDI

researchers field-level evidence for seasonal reporting. Because the architecture is cloud-based and the hardware low-cost, it can scale to many farms through cooperatives and institutional subscriptions.

Contribution to the Sustainable Development Goals

The component contributes mainly to SDG 2 (Zero Hunger) by supporting more stable smallholder production, and to SDG 13 (Climate Action) by helping farmers adapt to monsoon variability [19]. Through its links with the irrigation component it can also support SDG 6 (Clean Water and Sanitation) via more efficient water use, and through its low-cost sensing and open-source stack it relates to SDG 9 (Industry, Innovation and Infrastructure). These contributions are potential, not measured, outcomes of this thesis.

Scalability, compliance and sustainability

Scaling beyond the pilot requires more varieties and seasons in training data, sensor maintenance and replacement, retraining as new harvest data arrive, and compliance with data-protection requirements for farmer records.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis presented the Paddy Yield Forecasting component of an IoT and AI smart agriculture platform for Sri Lankan smallholder paddy farming. The component combines live soil and weather measurements with RRDI historical records for BG 352 and BG 367, uses an XGBoost model with monsoon and growth-stage features benchmarked against Random Forest and ANN baselines, and delivers four-to-six-week-ahead forecasts through APIs and a Sinhala/English interface.

8.2 Achievement of Objectives

Table 8.1 Achievement of the specific objectives

ID	Objective	Evidence
SO1	Collect soil, weather and RRDI data	Section 3.2; Section 5.2
SO2	Pre-processing and feature engineering	Sections 3.3 and 5.2
SO3	Train and benchmark XGBoost against baselines	Sections 6.1 and 7.2
SO4	Integrate predictions into app and APIs	Sections 4.4, 5.2 and 5.3
SO5	Validate against Yala 2026 harvest data	Section 6.3

8.3 Individual Research and Product Contributions

Research: an empirical comparison of soil-aware and weather-only forecasting for Sri Lankan paddy, and a benchmark of XGBoost against Random Forest and ANN with a validation design suited to field data. **Product:** a reproducible pipeline from sensor ingestion to API delivery, integrated into a bilingual farmer interface and shared platform.

8.4 Future Directions

- Extend training and validation to further varieties, to the Maha season and to other districts and agro-ecological zones.
- Collect multi-season data to increase the number of independent field observations and re-evaluate accuracy.

- Provide probabilistic forecasts (for example, prediction intervals) so that the confidence score is statistically calibrated.
- Fuse IoT data with satellite indices (for example NDVI) to combine plot-level and field-scale information.
- Evaluate temporal models (for example LSTM) on denser sensor time series.
- Run inference on edge devices for fully offline operation, and close the feedback loop by learning from farmer-reported harvests.
- Strengthen interpretability so that farmers and officers can see which conditions drive each forecast.

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GLOSSARY

BG 352 and BG 367 – Rice varieties bred in Sri Lanka and used in this study; historical records are held by RRDI.

Yala and Maha – The two main paddy cultivation seasons in Sri Lanka (Yala: roughly May–August; Maha: roughly September–March).

Electrical conductivity (EC) – A measure of the salt content of the soil solution, used here as a soil-condition indicator.

Growth stage – A phase of paddy development, such as tillering, panicle initiation, flowering or grain filling.

Feature engineering – Constructing informative model inputs from raw data, for example cumulative rainfall or growth-stage indicators.

Feature store – A store of computed features used consistently for training and inference.

Ablation test – An experiment that removes or replaces a group of inputs to measure its contribution to accuracy.

Cross-validation – Repeatedly training and testing a model on different splits of the data to estimate generalisation.

t/ha – Tonnes per hectare, the unit of paddy yield used in this thesis.

APPENDICES

Supplementary material cited in the thesis is provided in Appendices A and B.

Appendix A Data Fields and Extended Results

Table A.1 lists the data fields used by the component (schema only; the underlying RRDI data and farmer records are restricted).

Table A.1 Sensor and historical data fields

Data set	Field	Unit / type
Sensor	Timestamp, node ID, plot ID (anonymised)	Date-time; text
Sensor	Soil moisture	[% or volumetric water content]
Sensor	Electrical conductivity	[dS/m]
Sensor	Soil pH	pH units
Sensor	Soil/air temperature	°C
Sensor	Rainfall	mm
Sensor	Relative humidity	%
RRDI	Variety, season, year, location	BG 352 or BG 367; Yala/Maha; year; text
RRDI	Recorded yield	t/ha

Appendix B Academic Integrity and AI Use Disclosure

In accordance with the module policy, the use of generative AI is disclosed below. The author remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions of this thesis.

Item	Disclosure
Tool	Claude (Anthropic)
Purpose	Organising the author's own proposal report, final presentation and the thesis template into the template's chapter structure; drafting and editing prose; checking selected references against publisher records.
Inputs supplied	The project proposal (March 2026), the group final presentation and the individual thesis template.
Verification performed	The author-supplied entries for Amaratunga et al. and Ekanayake et al. were corrected after checking publisher records; additional standard references were added and must be checked by the author.
Sections affected	All chapters; no experimental results, data or code were generated by the tool. Result tables and bracketed items were left for the author to complete from the actual experiments.